LLM-6XX | Assignment 1

Urdu News Headline Classification with Parameter-Efficient Fine-Tuning

Stage 1 Track B Delivery Report
Infinitic AI coursework project

Group members

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Finalist model

xlm-roberta-base · full fine-tune · cosine schedule

Primary results

Validation macro-F1: **0.9481** | Test macro-F1: **0.9384**

Deadline: 20 October 2026

Repository: github.com/HammadHARahim/Infinitic-AI-Urdu-News-Headline-Classifier

1. Project description

Given an Urdu news headline, predict its news category. Urdu is a low-resource language written in a right-to-left Perso-Arabic script. This project measures how well a multilingual pretrained transformer handles that setting, how close LoRA gets to full fine-tuning, and which schedule or regularization choices improve validation macro-F1.

Figure 1. Stage 1 end-to-end pipeline

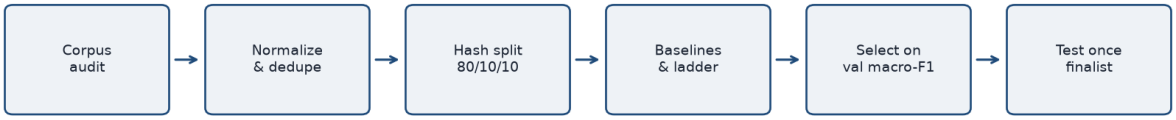


Figure 1. Stage 1 end-to-end pipeline.

Research questions

- RQ1: How close does LoRA get to full fine-tuning, and with how many trainable parameters?
- RQ2 (partial): How does the XLM-R tokenizer behave on Urdu headlines?
- RQ4 (partial): Do class weights, weight decay, and label smoothing help?

RQ3 (Qwen2.5 prompts) and E8–E10 (input length, other models, English-only BERT) remain after the deadline.

2. Dataset and preprocessing

Stage 1 uses the public **Urdu News Dataset 1M** (Kaggle copy of Mendeley v3, DOI 10.17632/834vsxnb99.3) for non-commercial research. Direct scraping of several Urdu news sites was blocked by terms of use.

Figure 6. Stage 1 label counts before the hash split

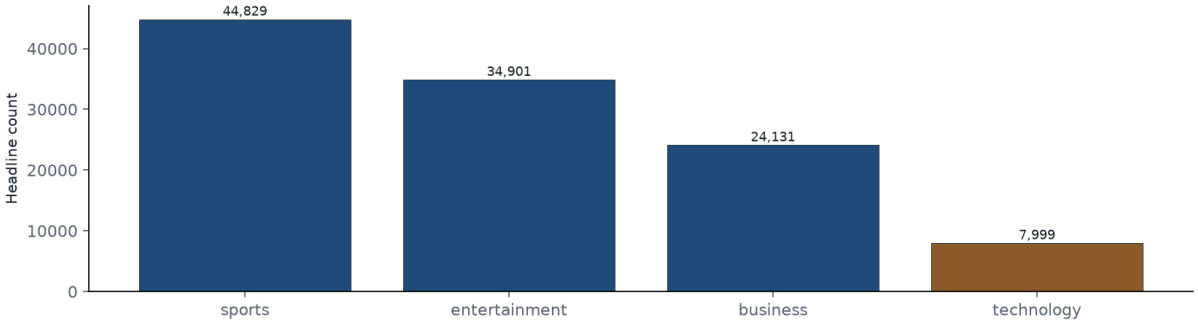


Figure 6. Stage 1 label counts (technology is the minority class).

Item	Value
Rows kept after audit	111,860
Stage 1 labels	sports, business, entertainment, technology
Split	80% train / 10% validation / 10% test · seed 42
Split method	Normalized-content hash; near-duplicates stay together
Test headlines	11,252 (sealed until finalist)

Table 1. Stage 1 corpus summary.

3. Model and training setup

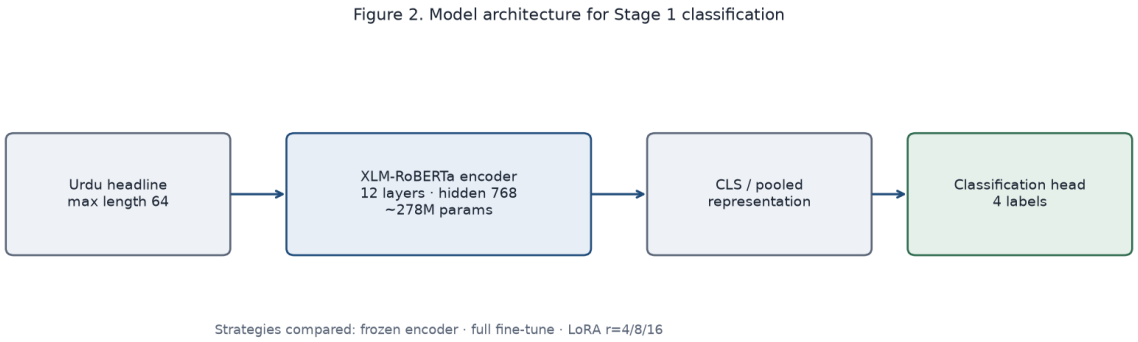


Figure 2. Model architecture for Stage 1 classification.

Main model: **xlm-roberta-base** (~278M parameters). Hardware: Kaggle Tesla T4, batch size 32, max length 64. Optimizer AdamW. Selection metric: validation macro-F1 only.

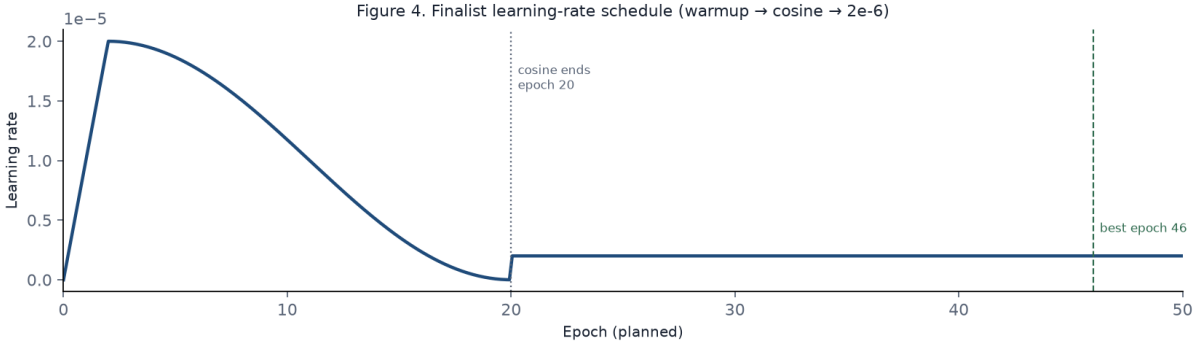


Figure 4. Finalist learning-rate schedule.

Setting	Value
Strategy	Full fine-tune (not LoRA)
Peak learning rate	2e-5
Schedule	10% warmup, half-cosine over 20 epochs, then constant 2e-6
Weight decay / loss	0.01 / unweighted cross-entropy
Epochs scored / best epoch	48 of planned 50 / epoch 46
Validation / test macro-F1	0.9481 / 0.9384
Checkpoint	HammadHARahim/urdu-headline-stage1-cosine

Table 2. Declared Stage 1 finalist configuration.

4. Experiment ladder

Each run changed one controlled variable. A setting joined the recipe only if it beat the kept validation macro-F1.

Figure 3. Validation macro-F1 across the Stage 1 ladder

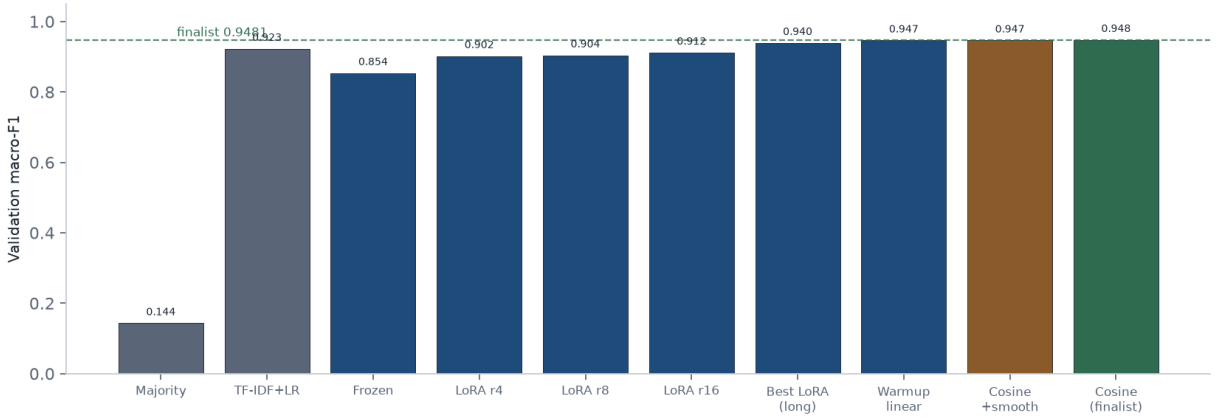


Figure 3. Validation macro-F1 across the Stage 1 ladder.

ID	Experiment	Status	Val macro-F1	Notes
—	Majority / TF-IDF+LR	Done	0.1437 / 0.9230	Baselines
E2	Frozen encoder	Done	0.8538	2 epochs
E3	Full fine-tune	Done	0.9481	Cosine finalist
E4a-c	LoRA r=4/8/16	Done	≤0.9395	Below full model
E5	WD / label smoothing	Partly	≤0.9466	Did not beat 0.9481
E6	Warmup / cosine	Done	0.9466 → 0.9481	Cosine won
E7	Class weights	Done	0.9430	Did not beat then-best
E1	Qwen prompts	Remaining	—	RQ3
E8–E10	Length / other models	Remaining	—	After deadline

Table 3. Assignment ladder coverage.

5. Main results

Run	Epochs scored	Best epoch	Val macro-F1
Warmup + linear decay	20	19	0.9466
Cosine + 2e-6 (finalist)	48	46	0.9481
Cosine + label smoothing	60	36	0.9466
Best LoRA (r=4 + smoothing)	100	56	0.9395

Table 4. Headline full-model and best LoRA scores.

- Full fine-tuning beat LoRA (0.9481 vs 0.9395).
- Cosine continuation beat linear warmup by +0.0015 validation macro-F1.
- Label smoothing, stronger weight decay, and class weights did not enter the recipe.

6. Final test evaluation (once)

On 20 October 2026 the kept cosine checkpoint was declared the Stage 1 finalist. A fresh retrain from base was skipped because Kaggle GPU quota was 0.00h. The sealed test split was opened once.

Metric	Score
Test macro-F1	0.9384
Test accuracy	0.9579

Metric	Score
Test size	11,252 headlines

Table 5. Finalist test summary.

Figure 5. Finalist test confusion matrix

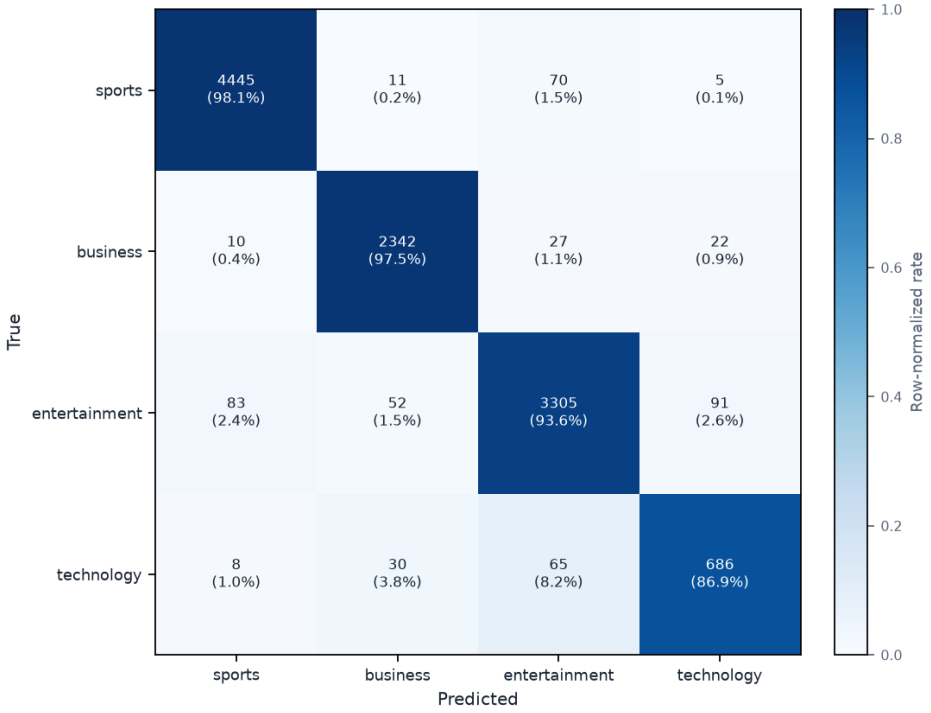


Figure 5. Finalist test confusion matrix (counts and row rates).

Label	Precision	Recall	F1	Support
sports	0.9778	0.9810	0.9794	4531
business	0.9618	0.9754	0.9686	2401
entertainment	0.9533	0.9360	0.9446	3531
technology	0.8532	0.8695	0.8613	789

Table 6. Per-label test metrics. Technology remains hardest.

7. Limitations and remaining work

- Four labels only in Stage 1; politics / health / world need another licensed corpus.
- E1 and E8–E10 were not completed before the deadline.
- Preferred fresh finalist retrain from base was not run (GPU quota).
- Track A (from-scratch Urdu transformer) is future work.

8. Conclusion

Stage 1 Track B delivered a reproducible fine-tuning ladder on xlm-roberta-base, selected the cosine-schedule full fine-tune on validation macro-F1 (0.9481), and scored the sealed test once (0.9384). LoRA saved parameters but did not match full fine-tuning. Schedule choice mattered more than label smoothing or class weights.

9. Roles and packaging

Member	Primary focus for this delivery
Hammad Rahim	Training ladder, Kaggle runs, finalist selection, repository
Sultan Tufail	Data audit / split support, experiment logging, review
Adil Abdul Haq	Evaluation, report packaging, remaining-experiment notes

Table 7. Group roles for Stage 1 delivery.

Artifacts: public GitHub repository, experiments/registry.csv, score reports under data/processed/, diagrams under docs/figures/, and Hub checkpoint HammadHARahim/urdu-headline-stage1-cosine.